



NORTH SYDNEY GIRLS HIGH SCHOOL

YEAR 12 – TERM 2 ASSESSMENT

2006

MATHEMATICS EXTENSION 1

TIME ALLOWED: One Hour
Plus 2 minutes reading time

INSTRUCTIONS:

- Start each question on a new page
- Hand each question in separately, including a sheet for non-attempts
- Show all necessary working

This task is worth 20% of the HSC Assessment Mark

Question One – (10 marks)

	Marks
a) Find the exact value of	
(i) $\sin^{-1}\left(\sin\frac{2\pi}{3}\right)$	1
(ii) $\tan\left(\cos^{-1}\frac{1}{3}\right)$	2
b) Find the following integrals	
(i) $\int \frac{dx}{\sqrt{16-x^2}}$	1
(ii) $\int \frac{dx}{4+3x^2}$	2
c) Two circles intersect at A and B . CAF and EBD are straight lines. Prove that CE is parallel to FD .	4

C

A

F

E

B

D

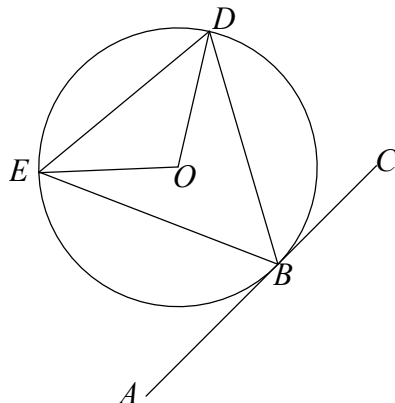
Question Two – (9 marks)

Point $P(2ap, ap^2)$ lies on the parabola $x^2 = 4ay$

- | | | |
|-------|--|---|
| (i) | Show that the equation of the tangent to the curve at P is
$y = px - ap^2$. | 2 |
| (ii) | This tangent cuts the x axis at T .
Find the coordinates of T . | 1 |
| (iii) | If S is the focus of the parabola prove that ST and PT are at right angles to each other. | 3 |
| (iv) | Show that the locus of the centre of the circle that passes through P , S and T is the curve $2ay = a^2 + x^2$. | 3 |

Question Three – (11 marks)**Marks**

- a) ABC is a tangent at B to the circle centre O . $\angle ABE = 50^\circ$ and $\angle BED = 65^\circ$. Find the size of $\angle DOE$ giving reasons for your answer. **3**



- b) Find the equation of the normal to the curve $y = \tan^{-1}(2x)$ at the point where $y = \frac{\pi}{4}$. **4**

- c) Find the derivative of $\sin^{-1}(x-1)$ and hence evaluate

$$\int_{\frac{1}{2}}^1 \frac{dx}{\sqrt{x(2-x)}} \quad \mathbf{4}$$

Question Four (10 marks)

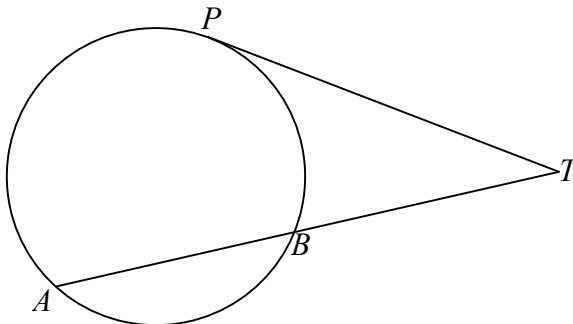
- a) Solve the equation $\sin x + \cos x = 0$ for all real x . **2**

- b) State the domain and range of $3y = \sin^{-1}\left(\frac{x}{2}\right)$ and sketch the curve. **3**

- c) Prove, by Mathematical Induction, that $\frac{2^n - (-1)^n}{3}$ is odd for all positive integers n . **5**

Question Five (10 marks)**Marks**

- a) PT is a tangent to the circle at P . $AB = 12\text{cm}$, $PT = 8\text{cm}$. Find the length of BT giving reasons for your answers.

3

- b) If $f(x) = \frac{x-4}{x-2}$, find $f^{-1}(x)$ and find its range.

4

- c) Show that $\tan^{-1}\left(\frac{1}{2}\right) - \tan^{-1}\left(\frac{1}{4}\right) = \tan^{-1}\left(\frac{2}{9}\right)$

3**Question Six (10 marks)**

- a) If $y = \frac{\cos^{-1}\left(\frac{x}{3}\right)}{x}$, find $\frac{dy}{dx}$.

3

- b) $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ are two points on the parabola $x^2 = 4ay$.

- i) Show that the equation of the chord PQ is $y - \left(\frac{p+q}{2}\right)x + apq = 0$.

2

- ii) If the chord PQ passes through the focus of the parabola show that $pq = -1$.

1

- iii) If M is the midpoint of the focal chord PQ , K is the foot of the perpendicular from M to the directrix and N is the midpoint of MK , find the equation of the locus of N .

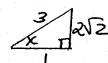
4**End of paper**

SOLUTIONS

QUESTION 1 10 MARKS

a) (i) $\frac{\pi}{3}$

(ii) Let $x = \cos^{-1}(\frac{1}{3}) \Rightarrow \cos x = \frac{1}{3}$



$\tan x = \sqrt{3}$

b) (i) $\sin^{-1}(\frac{x}{4}) + c$

(ii) $\int \frac{dx}{4+3x^2} = \frac{1}{3} \int \frac{dx}{\frac{4}{3} + x^2}$

$= \frac{1}{3} \times \frac{\sqrt{3}}{2} \tan^{-1} \frac{x}{\frac{2}{\sqrt{3}}} + c$

$= \frac{\sqrt{3}}{6} \tan^{-1} \frac{\sqrt{3}x}{2} + c$

c) Join AB

$\therefore$ CABE, AFDB are both cyclic quadrilaterals

Let $\angle ABD = x^\circ$

$\therefore \angle ECA = x^\circ$ (exterior angle of cyclic quadrilateral equals interior opposite angle)

$x + \angle AFD = 180^\circ$ (opposite angles of cyclic quadrilateral are supplementary)

$\therefore \angle AFD = (180 - x)^\circ$
 $\angle AFD + \angle ECA = (180 - x) + x$
 $= 180^\circ$

$\therefore CE \parallel FD$ (supplementary interior angles)

QUESTION 2 9 MARKS

(i) $x^2 = 4ay$

$y = \frac{x^2}{4a}$

$\frac{dy}{dx} = \frac{2x}{4a}$

when $x = 2ap$, $\frac{dy}{dx} = p$

$y - ap^2 = p(x - 2ap)$

$y - ap^2 = px - 2ap^2$

$y = px - ap^2$

(ii) $0 = px - ap^2$

$x = ap$

$T(ap, 0)$

$S(0, a)$

Slope ST = $\frac{a}{-ap}$

$= -\frac{1}{p}$

Slope PT = $\frac{ap^2}{2ap - ap}$

$= p$

$-\frac{1}{p} \times p = -1$

$\therefore ST \perp PT$

SOLUTIONS p 2

QUESTION 2 (cont)

(iv) Centre of circle is midpoint of PS

i.e. $(ap, \frac{ap^2+a}{2})$

$x = ap \Rightarrow p = \frac{x}{a}$

$2y = a \frac{x^2}{a^2} + a$

$2ay = x^2 + a^2$

QUESTION 3 11 MARKS

a) $\angle BDC = \angle BED$ (angle between tangent and chord at point of contact equals angle in alternate segment)

$\angle ABE = 50^\circ$ (given)

$\angle BDC + \angle ABE + \angle EBD = 180^\circ$ (angles form straight angle)

$\therefore \angle EBD = 180^\circ - (50 + 65)^\circ$

$= 65^\circ$

$\angle DOE = 2 \times \angle EBD$ (angle at centre is twice angle at circumference standing on same arc)

$= 130^\circ$

b) $y = \tan^{-1} 2x$

$\frac{dy}{dx} = \frac{2}{1+4x^2}$

when $y = \frac{\pi}{4}$, $x = \frac{1}{2}$, $\frac{dy}{dx} = 1$

$\therefore$ Slope of normal is -1

$y - \frac{\pi}{4} = -1(x - \frac{1}{2})$

$4y - \pi = -4x + 2$

$4x + 4y = \pi + 2$

c) $\frac{d}{dx} \sin^{-1}(x-1) = \frac{1}{\sqrt{1-(x-1)^2}}$

$= \frac{1}{\sqrt{1-x^2+2x-1}}$

$= \frac{1}{\sqrt{x(2-x)}}$

$\therefore \int \frac{dx}{\sqrt{x(2-x)}} = \left[\sin^{-1}(\frac{x-1}{1}) \right]_{\frac{1}{2}}^1$

$= \sin^{-1} 0 - \sin^{-1}(-\frac{1}{2})$

$= 0 - (-\frac{\pi}{6})$

$= \frac{\pi}{6}$

